

Billiards Match — Game Rules

Balls

Standard 8-Ball set (16 total): cue **cue** (1), solids **1-7** (7), 8 **8** (1), stripes **9-15** (7).

Objective

Clear your assigned group (solids or stripes), then legally pocket the 8 to win.

Assignment

Before each match, players are assigned solids or stripes. Targets:

- Solids: **['1', '2', '3', '4', '5', '6', '7']**
- Stripes: **['9', '10', '11', '12', '13', '14', '15']**

Shooting

- Legal shot: cue ball must first contact one of the shooter's target balls (or the 8 if targets are cleared).
- Pocketing your ball: continue shooting. Otherwise, turn passes.
- Fouls typically restore the table to the previous state and give the opponent the turn; certain fouls cause immediate loss.

Fouls (Key)

- Immediate loss: cue ball and 8 pocketed together; pocketing the 8 before clearing your group.
- Turn-change fouls: cue ball pocket (scratch), no contact, wrong first contact, or no cushion when no ball is pocketed.

Win / End Conditions

- Legal 8 after clearing your group → win.
- Immediate-loss foul → opponent wins.

- Max strokes: 60. Compare remaining balls (excluding 8); fewer remaining wins. Equal → SAME (draw).

Agents & Player Roles

- Agent A: BasicAgent . Agent B: LLMAgent .
- Player roles (first/second) rotate in 4-game cycles so each agent experiences all combinations of (first/second) × (solids/stripes).

Scoring

- AGENT_A_WIN / AGENT_B_WIN : 1 point.
- SAME : 0.5 point each.

$$\text{SCORE} = \text{WIN} + 0.5 \times \text{SAME}$$

$$\text{Rate} = \frac{\text{WIN} + 0.5 \times \text{SAME}}{N}$$

Execution Noise

The environment adds Gaussian noise to action parameters to simulate realism:

Parameter	Meaning	Std dev
v0	initial speed (m/s)	0.1
phi	horizontal angle (deg)	0.1
theta	vertical angle (deg)	0.1
a	lateral offset (ball-radius units)	0.003
b	longitudinal offset (ball-radius units)	0.003

Noise toggle: set enable_noise = True in poolenv.py (default).